Royco Dusk

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1 Overview

This section offers a brief primer on Royco's structuring substrates: Dawn and Dusk. Although this document specifies Dusk, the two substrates share their core machinery: tranches, the per-tranche NAVs, and the profit and loss waterfall.

1.1 Tranches and NAVs

A Royco market structures an asset into two complementary investment opportunities with distinct risk/return profiles called *tranches*:

- **Senior tranche (ST):** Protected capital. Lower returns for receiving first-loss coverage.
- **Junior tranche (JT):** Levered capital. Higher returns for providing first-loss coverage.

Moreover, each tranche holds a perpetual obligation to the other. The senior tranche pays the junior a share of its yield (risk premium) for absorbing losses first. The junior tranche, in turn, stands as a first-loss buffer: senior losses consume junior capital in full before the senior tranche incurs any. Accounting for these obligations requires distinguishing what each tranche *deposited* from what it is *entitled to*. Hence, Royco tracks two NAVs (net asset value) per tranche:

- **Raw NAV:** The mark-to-market value of the tranche's *deposited* assets.
- **Effective NAV:** The value that each tranche *is entitled to* after its obligations to the other are settled.

1.2 The profit and loss waterfall

Every accounting synchronization marks the tranche NAVs to market in three steps:

1. **Measure:** Read the current raw NAVs and compute the delta against their last checkpointed values. Each delta represents the unrealized profit or loss of that tranche's holdings since the previous accounting sync.
2. **Attribute:** Past coverage and risk premiums can leave a tranche entitled to more than its own holdings are worth. The excess ($ST_{\text{eff}} - ST_{\text{raw}}$ when positive, and symmetrically for JT) is a claim on the *other* tranche's holdings. By conservation the two excesses are equal and opposite, so one tranche's excess is exactly the other's shortfall and the cross-claim runs one way. Each side's raw NAV delta is split pro rata over the claims on it. The result is a pair of effective NAV deltas that sum to the total profit and loss.
3. **Settle:** The effective deltas are routed through the *PnL waterfall* in fixed priority order:
   - *Loss:* Utilize JT's coverage buffer (JT_{eff}) before ST books any. Any coverage that JT advances to ST is booked as *JT impermanent loss* (ST's debt to JT). Losses exceeding JT's buffer reduce ST_{eff} and are booked as *ST impermanent loss* (JT's debt to ST). JT IL is transient: it exists only during the market's fixed-term regime, a window that lets the tranched asset recover before the junior realizes losses.
   - *Recovery:* Repay any debts before gains accrue. ST's IL has the first claim on appreciation from either side, and JT's IL has the second claim on senior appreciation only.
   - *Yield:* Junior appreciation remaining after ST IL repayment accrues to JT in full. Senior appreciation remaining after all IL repayment is true yield, split between ST and JT at the rate set by the yield distribution model (YDM).

We denote the attribution and settlement steps as F throughout this document. Holding the checkpoint fixed, F maps the current raw NAVs to the new effective NAVs:

$$F\colon (ST_{\text{raw}},\ JT_{\text{raw}};\ \text{checkpoint}) \longmapsto (ST_{\text{eff}},\ JT_{\text{eff}}).$$

The semicolon separates variables from parameters: fixing the checkpoint yields a function of the raw NAVs alone. The checkpoint holds the previous sync's raw NAVs, effective NAVs, outstanding ILs, market state, and yield-accrual bookkeeping (the time-weighted YDM accumulator and its timestamps). It is read from storage when the sync begins and cannot change while it runs. Every property stated for F holds for each fixed checkpoint.

F holds two key properties:

1. **NAV conservation:** the inputs sum to the outputs, $ST_{\text{raw}} + JT_{\text{raw}} = ST_{\text{eff}} + JT_{\text{eff}}$, for every checkpoint that itself conserves. Every stored checkpoint conserves, because it is itself either the conserving genesis checkpoint seeded at market creation or a prior output of F. F redistributes value between the tranches without creating or destroying it.
2. **Piecewise-affine:** F is built from chains of min, max, and affine pieces with slopes $\in [0, 1]$, so it is continuous and non-decreasing in each input. NAV conservation then makes it 1-Lipschitz in each input. The existence proof invokes the junior direction (Section 4).

1.3 Dawn to Dusk

Dawn transforms the *risk profile* of any asset, but not its *liquidity profile*: senior shares have no exit path other than primary redemption.

Dusk extends the junior tranche to transform both. In Dawn, junior capital may sit in any priceable asset (Treasuries, a money market fund, or co-investment alongside the senior tranche), provided its value is exogenous to the market itself. Dusk drops that restriction: junior capital is deployed into an AMM pool paired against the senior tranche's shares.

- **Risk transformation:** Coverage still flows from JT to ST exactly as in Dawn: the pool position is JT's first-loss collateral.
- **Liquidity transformation:** The pool position gives senior holders an instant secondary market, constantly quoting a price to buy and sell tranche shares for stables.

For the purposes of this document, the Dusk JT's asset is a Balancer V3 Pool Token (BPT) whose constituents are the senior tranche share and a stable quote asset (USDC, sUSDS, or srRoyUSDC).

The recursion: Marking JT to market is no longer a plain oracle read. To value the BPT, the pool's senior share leg must be priced at the senior rate $y = ST_{\text{eff}}/N$, where N is the total supply of senior shares. But ST_{eff} is not an observable quantity: it is the output of the sync map F, and F cannot run without its JT_{raw} input, which is the BPT value we set out to compute. The dependency chain closes on itself, as Figure 1 shows.

Economically, the junior tranche's collateral contains the very senior shares its coverage protects. No evaluation order can resolve the cycle: the two unknowns, JT_{raw} and ST_{eff} (y is simply ST_{eff} per share), must be determined simultaneously. Under conditions made precise in Section 4, the values of ST_{eff} that satisfy both dependencies at once form a contiguous block: price the BPT at its implied rate $y = ST_{\text{eff}}/N$, apply F, and the same ST_{eff} returns. Each such self-reproducing value is a *fixed point*, and the published value is the largest. Section 3 formulates the equation, Section 4 proves the published value exists and is uniquely selected, and Section 5 finds it on-chain.

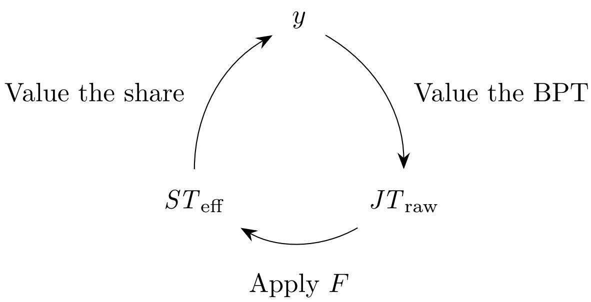

Figure 1: The recursion. Valuing the junior collateral needs the senior rate, and the senior rate depends on that valuation.

1.4 Components of a Dusk market

A Dusk market adds three components to the Dawn stack (pool, kernel, hook) and reuses the Dawn accountant. Each plays one role in resolving the recursion:

- **Gyro E-CLP pool:** A Balancer V3 elliptic concentrated liquidity pool over the senior share and quote asset. The senior leg is priced by the rate the kernel publishes, so the solved rate is what the pool uses to value senior shares.
- **Dusk kernel:** One contract, two paths. Write: the sync entrypoint solves the fixed point (Section 5), commits the accountant checkpoint, and caches the solved rate $y^\star$. Read: the kernel reports the cached rate $y^\star$ to the pool instantly.
- **Balancer V3 Dusk hook:** Forwards Balancer's before-operation callbacks (swap, add and remove liquidity) into the kernel's sync entrypoint. Balancer reloads balances and rates after the hook so every standard operation executes at a fresh senior share rate.
- **Accountant:** Shared with Dawn. During a solve, the kernel calls its view-only preview repeatedly to evaluate F at candidate rates. After convergence, exactly one call commits the resultant checkpoint.

2 Notation

This section defines the notation used in subsequent sections.

Unless stated otherwise, every quantity is an 18-decimal fixed-point integer (WAD, scale 10^{18}): NAV amounts, rates, ratios, senior tranche shares, and BPT amounts alike. The single exception is raw quote-asset amounts, which keep their token's native decimals (e.g. 10^6 for USDC).

Table 1: Mathematical notation used throughout this document.

Symbol	**Meaning**	**Units**
Tranche NAVs and supply		
ST_{raw}	Raw NAV of the senior tranche's own assets	NAV
JT_{raw}	Raw NAV of the junior tranche's BPT collateral, $JT_{\text{raw}} = P_A(y)$	NAV
ST_{eff}	Senior effective NAV after the waterfall	NAV
JT_{eff}	Junior effective NAV after the waterfall	NAV
y	Senior rate (NAV per senior share), the candidate $y = x/N$ fed into the solve	NAV/share
$y^\star$	Published senior rate, $y^\star = x^\star/N$ rounded toward senior, $= ST_{\text{eff}}/N$ at the largest fixed point	NAV/share
y_0	Cached senior rate, read once at solve start (fixes L and the band)	NAV/share

Symbol	Meaning	Units
x	Candidate senior NAV (the bisection probe), self-consistent exactly where $E(x) = 0$. The largest such point is the published $x^\star$	NAV
N	Total senior-share supply (including any fee shares), frozen at solve start. The same N appears in the clamp and the rate	shares
The fixed-point machinery		
F	Sync map, $(ST_{\text{raw}},\ JT_{\text{raw}};\ \text{checkpoint}) \mapsto (ST_{\text{eff}},\ JT_{\text{eff}})$, read as $JT_{\text{raw}} \mapsto ST_{\text{eff}}$ once the rest is fixed (Section 3.2)	function
checkpoint	Previous sync's raw NAVs, effective NAVs, outstanding ILs, market state, and yield-accrual bookkeeping	—
P_A	Conservative BPT valuation, $JT_{\text{raw}} = P_A(y)$	function
G	Return map (loop output), $G(x) = F(P_A(x/N))$, so $E(x) = G(x) - x$	function
E	Error function, $E(x) = F(P_A(x/N)) - x$, the bisection target	function
g	Loop gain (round-trip amplification of x). The clamp at N gives $g = f$, and $f \leq 1$ since the junior owns at most the whole pool	dimensionless
Pool reserves and geometry (Gyro E-CLP)		
b_q	Quote-leg reserve at the fair point	NAV
q	Quote-asset rate (1 for non-rate-bearing quote tokens)	NAV/quote
f	JT's fractional share of the total BPT supply	dimensionless
α, β	Per-solve lower / upper bounds on the senior rate the pool can quote, derived from the configured E-CLP band (Section 5.3, distinct from Royco's correlation parameter)	NAV/share
L	E-CLP invariant value	—
$V_{\max}$	Pool value at the all-quote corner of the frozen curve, $V_{\max} = \sup_y V(y)$	NAV
$V(y)$	Fair pool value at marginal price y (envelope-theorem valuation)	NAV
$D(y)$	The pool's demand for senior shares at price y (the fair reserve), $D(y) = V'(y)$	shares

3 The fixed-point formulation

Every standard pool operation triggers a sync, and every sync must publish one number: the senior share rate y. Section 1.3 showed why this is a fixed point, not an oracle read. This section makes it precise: one equation in one unknown, then two facts about one function.

3.1 What we must prove

Write the senior NAV the sync must produce as x. We will see that a self-consistent x is exactly a zero of the *error function*

$$E(x) \;=\; F(P_A(x/N)) - x,$$

where P_A values the junior collateral at a candidate rate and F is the accountant's sync map. The on-chain solver finds the largest such zero by integer bisection, and the argument needs exactly two facts:

(Monotonicity) E never rises: each step holds or falls by exactly one. So the values where $E \geq 0$ form an initial segment whose right endpoint, the crossing, is *unique* even when several zeros tie along it, and bisection knows which way to step.

(Bracketing) E is non-negative at the bottom of the search range and negative at the top. This makes the crossing *exist* and brackets it.

The rest of this section defines the three objects F, P_A, and E. Section 4 proves the two facts. Section 5 runs the solver. The argument rests on nothing else.

3.2 The sync map F

On each sync, the Dusk kernel calls the shared Royco accountant, whose sync map we denote F. Given the current raw NAVs of both tranches, F runs the attribution and settlement steps of Section 1.2 against the stored checkpoint and returns the effective NAVs:

$$F\colon (ST_{\mathrm{raw}},\ JT_{\mathrm{raw}};\ \text{checkpoint}) \longmapsto (ST_{\mathrm{eff}},\ JT_{\mathrm{eff}}).$$

Two properties of F matter downstream (Section 1.2). It conserves NAV ($ST_{\mathrm{raw}} + JT_{\mathrm{raw}} = ST_{\mathrm{eff}} + JT_{\mathrm{eff}}$), and it is non-decreasing and 1-Lipschitz in JT_{raw} (Lemma 4.2). Conservation holds by construction: every stored checkpoint is itself a prior output of F.

In a Dusk market the senior raw NAV is exogenous to the loop: it is the value of the senior tranche's own assets, marked independently of the pool. Only the junior raw NAV depends on the senior rate, because the junior's collateral is the BPT. The senior side therefore drops out, and for one sync, with ST_{raw} and the checkpoint fixed, we read F as a function of a single variable:

$$F\colon JT_{\mathrm{raw}} \longmapsto ST_{\mathrm{eff}}.$$

F returns ST_{eff} *gross* of protocol fees: a fee accrues as a separate NAV claim settled outside F, never netted out of ST_{eff}. So $y = ST_{\mathrm{eff}}/N$ is consistent only when N counts the same shares the gross NAV backs, including any minted for fees.

3.3 The BPT valuation P_A

The junior tranche owns a fraction f of a Gyro E-CLP pool whose two legs are the senior share and a quote asset. $P_A(y)$ answers one question: with the senior share priced at a candidate rate y, what is that stake worth? The stake is the junior's first-loss collateral, so the mark must never overstate it: coverage credited against an inflated mark is coverage the senior cannot collect. Three ingredients keep the mark honest: a curve frozen at solve start, a fair point on that curve, and a cap at the total share supply.

The frozen curve. Live pool balances can be pushed around within a block, so the mark never reads them directly. At the start of a solve the kernel reads the cached rate y_0 once and computes the pool invariant L from the rate-scaled balances, rounded down in favour of the senior tranche. L is then held fixed while the candidate rate y varies, which makes P_A a function of y alone. The monotonicity proof relies on exactly this. The scaling and rounding details are collected in Section 5.7.

The fair point. Given L, the pool is valued at the one point on the frozen curve consistent with the candidate price: the point whose *marginal price* (its local slope, the spot rate for an infinitesimal trade) equals y. Valuing any other point would mark the pool as if it were quoting a price the candidate rate contradicts. Write $V(y)$ for the pool's fair value at that point, and

$$D(y) \;=\; V'(y),$$

the pool's *demand* for senior shares at price y, that is, the senior reserve the fair point holds. The proofs need one geometric fact, the envelope theorem of Section 4.1: V is non-decreasing, with $V'(y) = D(y) \geq 0$ (a higher senior price raises the value of the senior shares the pool holds). The structural facts an auditor confirms per market are collected in Section 4.3.

The clamp. At low rates the frozen curve can demand more senior shares than exist: the quote leg would notionally absorb more than the total supply N, even if every holder sold into the pool. Coverage resting on shares beyond the N that exist is phantom coverage, so the demand is capped at the supply:

$$\widehat{D}(y) \;=\; \min\bigl(D(y),\ N\bigr), \qquad \widehat{V}(y) \;=\; \int_0^y \widehat{D}(u)\,du.$$

By definition $\widehat{V}(0) = 0$, and with $\widehat{D} \le D = V'$ this gives $0 \le \widehat{V} \le V$ everywhere: the cap only ever lowers the mark. The two values differ by exactly the phantom area,

$$V(y) - \widehat{V}(y) \;=\; V(0) \;+\; \int_0^y \bigl(D(u) - N\bigr)^+\,du.$$

For the E-CLP the all-senior corner holds no quote, so $V(0) = 0$ and the gap vanishes at any rate below which demand never exceeds N. At healthy rates the cap is therefore inactive and $\widehat{V} = V$ there. Its bite is reserved for the danger zone.

The valuation. The junior tranche's pro-rata claim, rounded down in favour of the senior tranche, is the *conservative BPT valuation*:

$$P_A(y) \;=\; \lfloor\, f \cdot \widehat{V}(y)\, \rfloor. \tag{1}$$

By the geometric fact and the clamp, $f\widehat{V}$ is non-decreasing in y and Lipschitz with constant fN, so the floor moves P_A by 0 or 1 per unit step of x (Section 4.1).

Why trading cannot lift the mark. Within a solve, V depends only on L and y, never on the live pool composition. A fee-free swap conserves the pool's true invariant, and the L the solver consumes is a round-down underestimate of it, so $L \le L_{\mathrm{true}}$ for every reachable composition. A swap (or a donation, which Balancer V3 ignores since it accounts internally) can therefore move the mark only within the round-down band, never above the true fair value.

In summary, *conservative* carries three independent senses in this document, and the valuation is built to have all three.

Manipulation-bounded. The mark reads the frozen invariant L and the fair point, not live pool balances, so trading moves it only within the round-down band above.

Senior-favouring. Every rounding step (the floor in P_A, the round-down of L, the published rate toward senior) breaks ties for the protected tranche.

Capped lower bound. The clamp $\min(D, N)$ under-marks by exactly the phantom area above, precisely where demand exceeds the shares that exist.

The published senior rate $y^\star$ is built on this capped lower-bound valuation by design. Section 4 turns the same cap into unconditional monotonicity.

3.4 The fixed-point equation

Chaining the definitions, $JT_{\mathrm{raw}} = P_A(ST_{\mathrm{eff}}/N)$ and $ST_{\mathrm{eff}} = F(JT_{\mathrm{raw}})$, so the senior NAV must reproduce itself:

$$x \;=\; F\bigl(P_A(x/N)\bigr)\,. \tag{2}$$

A solution is a zero of the error function E (Section 3.1), and the published rate is $y^\star = x^\star/N$. Section 4 proves the zeros of E form one contiguous block whose largest element is the published root $x^\star$ (Monotonicity), inside a range across which E changes sign (Bracketing).

4 Existence and unique selection of the solved rate

This section proves the two facts from Section 3.1: E is non-increasing (Monotonicity), and its sign changes across a search range whose endpoints we can write down (Bracketing). Together they give a unique largest self-consistent rate, which bisection finds deterministically.

Theorem 4.1 (the solved rate exists and is uniquely selected)**.** *Fix $ST_{\text{raw}} \geq 0$, the checkpoint, the supply $N \geq 1$, the junior share $f \leq 1$, and the frozen invariant L at the start of a sync. On the bracket*

$$\mathcal{B} \;=\; \bigl[\,0,\; x_{\max}\,\bigr], \qquad x_{\max} \;=\; ST_{\text{raw}} + \lfloor f\,V_{\max} \rfloor + 1,$$

where $V_{\max} = \sup_y V(y)$ is the pool value at the all-quote corner (the senior shares all sold), the integer error function $E(x) = F(P_A(x/N)) - x$ is non-increasing with steps in $\{-1, 0\}$ (Monotonicity) *and satisfies $E(0) \geq 0$ and $E(x_{\max}) < 0$* (Bracketing)*. Hence the integers where $E \geq 0$ form an initial segment $[0, x^\star]$ with a unique right endpoint $x^\star$, the largest self-consistent senior NAV, and bisection on the predicate $E(x) \geq 0$ finds it.*

Sections 4.1 and 4.2 prove the two pillars, and Section 4.3 records what the argument assumes about the pool, so an auditor can check it per market.

4.1 Monotonicity: E is non-increasing

$E(x) = F(P_A(x/N)) - x$. The $-x$ term falls by one per step, so it is enough to show the composition $x \mapsto F(P_A(x/N))$ rises by at most one per step. We bound each stage in turn.

Stage 1: $f\widehat{V}$ moves by at most 1 per step, so P_A moves by 0 or 1. Fix the invariant L at the pre-solve cached rate, as Section 3.3 requires. The fair pool value at rate y on the frozen curve is the quote it holds plus its senior shares marked at y,

$$V(y) \;=\; b_q(y) \;+\; y\,D(y),$$

where $D(y)$ is the pool's demand for senior shares at price y and $b_q(y)$ is the quote reserve at the same fair point. Moving along the curve $db_q/dD = -y$ (the slope of the reserve curve is the price), so the rebalancing terms cancel:

$$\frac{dV}{dy} \;=\; \underbrace{\frac{db_q}{dD}\frac{dD}{dy} + y\,\frac{dD}{dy}}_{=\,0} + D(y) \;=\; D(y) \;\geq\; 0.$$

This is the envelope theorem: the first-order effect of the rebalancing variable vanishes at the fair point, leaving the direct term. The cancellation holds on the open band (α, β). Below it the pool is all senior and $V(y) = y\,D(\alpha)$ rises linearly. Above it the pool is all quote and V is constant at $V_{\max}$. Both pieces are non-decreasing and join the band continuously, so V is non-decreasing throughout.

The clamp now does its one job. With $\widehat{D}(y) = \min(D(y), N)$ the capped value $\widehat{V}$ has $\widehat{V}'(y) = \widehat{D}(y) \leq N$ everywhere, so for $y > y'$,

$$0 \;\leq\; \widehat{V}(y) - \widehat{V}(y') \;\leq\; N\,(y - y').$$

The candidate rate is $y = x/N$, so a unit step in x moves $f\widehat{V}$ by at most $fN \cdot \frac{1}{N} = f \leq 1$. A real increase of at most 1 changes a floor by 0 or 1, so $P_A = \lfloor f\widehat{V} \rfloor$ rises by 0 or 1 per step, and never falls.

Stage 2: F is 1-Lipschitz. This is Lemma 4.2 below: $0 \leq F(j) - F(j') \leq j - j'$ for $j > j'$.

Composition. The chain $x \mapsto x/N \mapsto P_A \mapsto F$ multiplies the per-stage factors,

$$\underbrace{\frac{1}{N}}_{\div N} \cdot \underbrace{fN}_{P_A} \cdot \underbrace{1}_{F} \;=\; f \;\leq\; 1,$$

since the junior owns at most the whole pool ($f \leq 1$). This product is the *loop gain* $g = f$: the dollars that return to the senior NAV after one trip around the loop, per dollar sent in. Capping demand at the supply holds it at or below one *by construction*, with no runtime condition to check.

On the integer lattice this gives the result directly. A unit step in x moves P_A by 0 or 1 (Stage 1), and F is 1-Lipschitz (Stage 2), so $F(P_A(x/N))$ rises by at most 1. Subtracting the $-x$ term,

$$E(x{+}1) - E(x) \;\in\; \{-1,\; 0\}.$$

E is non-increasing, so the integers with $E \geq 0$ form an initial segment $[0, x^\star]$. At its right endpoint the next step must be a drop of 1 (a flat step would keep $E \geq 0$), so $x^\star$ is the unique integer with $E(x^\star) \geq 0 > E(x^\star{+}1)$. The drop being exactly 1 and E integer-valued force $E(x^\star) = 0$, an exact fixed point. Bisection returns this $x^\star$, the largest self-consistent senior NAV. This proves Monotonicity.

Lemma 4.2 (F is non-decreasing and 1-Lipschitz in JT_{raw})**.** *Fix ST_{raw} and the checkpoint and vary $j = JT_{\text{raw}}$. Then $0 \leq F(j) - F(j') \leq j - j'$ for $j > j'$.*

Proof. F reaches ST_{eff} through two stages (Section 1.2), each with non-negative coefficients $\in [0, 1]$: attribution splits the raw delta by a claim fraction $a \in [0, 1]$ fixed at the checkpoint (each tranche's claim on a leg is at most that leg's raw NAV, by conservation and non-negative effective NAVs, so $a \leq 1$), and settlement routes the result through chains of min, max, addition and subtraction (slopes $\{0, 1\}$) and a yield split (fractions w, $1 - w$ with w capped). So ST_{eff} is non-decreasing in j, and so is JT_{eff}. NAV conservation finishes it: $ST_{\text{eff}} + JT_{\text{eff}} = ST_{\text{raw}} + j$, so the two slopes sum to exactly 1, and since both are non-negative each lies in $[0, 1]$. Thus F is non-decreasing and 1-Lipschitz. □

Remark 4.3 (The zero plateau and the loop gain)*.* The clamp fixes the loop gain at $g = f \leq 1$ (Section 4.1), which makes E non-increasing but permits it to be flat. Flatness can occur at zero whenever a floor step of P_A passes through an F-branch of slope one, so the zero set $\{x : E(x) = 0\}$ is a contiguous integer block at any gain, a *plateau* of self-consistent NAVs whenever it holds more than one point. The gain bounds its length. Each step inside the block holds E at 0, so $F(P_A(x/N))$ rises by exactly 1, and since F is 1-Lipschitz and P_A steps by 0 or 1 (Section 4.1), P_A rises by exactly 1 with it. A block of $k + 1$ zeros thus forces $f\widehat{V}$ across k integer boundaries in k steps of at most f each. The start lies strictly less than one unit above its floor, so crossing k boundaries requires a total rise strictly above $k - 1$, while the rise is at most kf. Hence $k(1 - f) < 1$, and for $f < 1$ the block holds at most $\lceil 1/(1 - f) \rceil$ integers. At $f = 1$ the bound lapses and the plateau is bounded only by the bracket itself. In real arithmetic, with the floor in P_A removed and F read along its affine branches (Section 1.2), E is continuous and falls by at least $1 - f$ per unit of x, so for every $f < 1$ it has a unique real root. Removing the floor raises E by less than one, so that root lies in $[x^\star,\; x^\star + 1)$ and the published $x^\star$ is the nearest lattice point from below. What $f < 1$ buys is a plateau bounded independently of the bracket and a real-arithmetic anchor, not a singleton zero set. The clamp alone supplies neither.

The plateau costs nothing, because the published value is *defined* to be the largest plateau integer, the $x^\star$ of Lemma 4.1, and that choice is immaterial to the junior tranche. Across the block $E = 0$ gives $ST_{\text{eff}} = x$, so by NAV conservation $JT_{\text{eff}} = ST_{\text{raw}} + P_A(x/N) - x$. A unit step inside the block forces P_A up by exactly 1 (the step inference above), whence

$$JT_{\text{eff}}(x{+}1) - JT_{\text{eff}}(x) \;=\; \big(P_A((x{+}1)/N) - P_A(x/N)\big) - 1 \;=\; 1 - 1 \;=\; 0.$$

The junior's entitlement is identical at every plateau point. This invariance is exactly as strong as F's integer conservation: it relies on $ST_{\text{eff}} + JT_{\text{eff}} = ST_{\text{raw}} + JT_{\text{raw}}$ holding at wei precision, the two outputs not independently rounded, which the accountant guarantees by construction (Section 1.2). The senior's entitlement is not invariant: $ST_{\text{eff}} = x$ rises one wei per plateau point, so self-consistent publications can differ by up to $\lceil 1/(1-f) \rceil - 1$ wei when $f < 1$, and at $f = 1$ only the bracket bounds the spread. Publishing the largest is one more tie broken for the protected tranche (Section 3.3). Existence does not degrade at $f = 1$: the $+1$ slack in $x_{\max}$ (Section 4.2) makes $E(x_{\max}) \le -1 < 0$ whatever the gain, so the bracket still straddles a crossing. A bracket-independent bound on the plateau needs $f < 1$. Safety needs only the clamp.

4.2 Bracketing: the endpoints straddle the sign change

We exhibit a range whose endpoints have opposite signs by construction, so no runtime sign-change check can fail.

Lower endpoint. At $x = 0$, $E(0) = F(P_A(0)) = ST_{\text{eff}} \ge 0$, since the accountant holds effective NAVs non-negative, whatever the junior collateral is worth. This consumes $ST_{\text{raw}} \ge 0$: at $x = 0$ the conserved total is $ST_{\text{raw}} + P_A(0) = ST_{\text{raw}}$, so non-negative outputs are consistent with conservation only when it holds (it also keeps the bracket well-formed, $x_{\max} \ge 1$).

Upper endpoint. P_A is bounded above by $\lfloor f V_{\max} \rfloor$. Since V is non-decreasing, $\widehat{V} \le V \le V_{\max}$ for every candidate rate, whichever way the pool sorts its tokens. By NAV conservation $ST_{\text{eff}} = ST_{\text{raw}} + JT_{\text{raw}} - JT_{\text{eff}} \le ST_{\text{raw}} + JT_{\text{raw}} \le ST_{\text{raw}} + \lfloor f V_{\max} \rfloor$ (using $JT_{\text{eff}} \ge 0$) for every candidate. At $x_{\max} = ST_{\text{raw}} + \lfloor f V_{\max} \rfloor + 1$ we therefore have $E(x_{\max}) = ST_{\text{eff}} - x_{\max} \le -1 < 0$.

Both endpoints are computed once, at solve start, from frozen values. The sign change is a theorem, not a checked precondition.

4.3 What the argument assumes about the pool

The proof rests on structural facts an auditor verifies once per market:

Waterfall. F is non-decreasing and 1-Lipschitz in JT_{raw} (Lemma 4.2), conserves NAV at wei precision with the two outputs not independently rounded, and returns non-negative effective NAVs. Bracketing consumes the last two directly (Section 4.2), and Lemma 4.2's proof consumes them together with the yield split fraction $\in [0, 1]$. These are properties of the shared accountant and its yield model, and hold for any Royco market using the standard waterfall.

Pool. The fair value V is non-decreasing in y, with $V' = D \ge 0$ by the envelope theorem (Section 4.1), the demand D is non-increasing, and the all-quote corner value $V_{\max}$ is finite. Monotone V holds for any constant-function pool. The finite corner is a property of the E-CLP's bounded band, and it is what makes the bracket endpoint $x_{\max}$ well-defined.

The proof never differentiates F, so it survives the waterfall's kinks, and never inverts a pool invariant, so it transfers unchanged across pool types: only the demand D that the clamp caps is pool-specific.

Out of scope. The solve fixes a manipulation-bounded rate. It does not monitor the live pool's *composition*. As senior shares accumulate and quote drains, the published mark stays correct, since it reads the frozen curve rather than live balances, but tracking that drift is a separate operational concern. So is Royco's coverage and correlation parameter, a distinct quantity from the pool's upper price bound β (Table 1). Neither is addressed by the fixed point.

5 The on-chain solver

Lemma 4.1 gives a unique largest root $x^\star$ in a bracket whose endpoints straddle the sign change. This section runs the solver: integer bisection, in bounded gas. The conditions that make the theorem apply on-chain are stated as explicit obligations.

5.1 Integer bisection

The solver bisects E on $\mathcal{B}$ at integer (WAD) precision. Each step halves the bracket and keeps the half whose endpoints still straddle the sign change. The bracket holds $|\mathcal{B}| = x_{\max}+1$ integers, so $\lceil \log_2(|\mathcal{B}|) \rceil$ steps collapse it to a single integer $x^\star$, the boundary with $E(x^\star) \geq 0 > E(x^\star+1)$. The deployed iteration count K is hardcoded per market as an upper bound on this count over every reachable bracket ($2^K \geq x_{\max}+1$, sized from the maximum representable $x_{\max}$), so the collapse completes within K steps on every solve. The published rate is $y^\star = x^\star/N$, rounded toward the senior tranche.

Bisection, not Newton, for three reasons. F is non-smooth, its slope jumping at branch boundaries (Lemma 4.2), so a tangent-following method has no derivative to follow, whereas bisection needs only that E be non-increasing. The iteration count K is fixed at deploy time, so gas is constant with no data-dependent exit. And the loop is a short sweep over already-audited valuation code, with no invariant inversion and a unique predicate boundary.

5.2 Probe value reuse

Write the loop output as $G(x) = F(P_A(x/N))$, so $E(x) = G(x) - x$. To get the sign of $E(z)$ the solver already evaluates $G(z)$ in full — it prices the junior collateral at the candidate rate and runs the waterfall — then keeps one bit and discards the number. But that number is not merely a test statistic. Monotonicity (Section 4.1) certifies $G(z)$ as a bracket endpoint on the correct side of the root, usually far tighter than the midpoint rule reaches in many steps.

Lemma 5.1 (probe upgrade)**.** *Let z be a probe and $v = E(z)$, so the loop output $G(z) = z + v$ is already in hand. If $v \geq 0$ then $z \leq x^\star$ and $G(z) \in [z,\ x^\star]$; if $v < 0$ then $z > x^\star$ and $G(z) \in [x^\star,\ z-1]$. Either way $G(z)$ is a valid endpoint on z's side of the root, no farther from $x^\star$ than z.*

Proof. By Monotonicity (Section 4.1) the integers with $E \geq 0$ are exactly $[0, x^\star]$, so the sign of v locates z relative to $x^\star$. And $G = F \circ P_A$ is non-decreasing, both factors being so (Section 4.1 and Lemma 4.2). If $z \leq x^\star$: $G(z) \leq G(x^\star) = x^\star$ and $G(z) = z + v \geq z$. If $z > x^\star$: $G(z) \geq G(x^\star) = x^\star$ and $G(z) = z + v \leq z - 1$, since $v \leq -1$. □

The solver applies the lemma at every step: where plain bisection moves the endpoint to the midpoint m, value reuse moves it to the value the midpoint returned, $G(m) = m + v$.

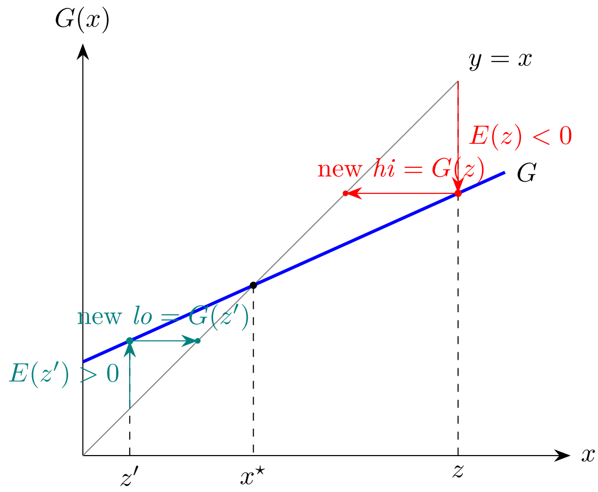

Figure 2: Probe value reuse, one step. The root $x^\star$ is where the return map G meets the diagonal $y = x$. A probe at z right of the root lands the curve point $(z, G(z))$ below the diagonal yet no lower than $x^\star$ (Lemma 5.1); sliding horizontally to the diagonal makes $G(z)$ the new upper endpoint, already past the midpoint and toward the root. A probe z' left of the root gives a new lower endpoint symmetrically. On a flat stretch of G — a covered, locally inert market state — the first slide lands at the root's level at once: the 2–3 probe solve of Lemma 5.2.

Maintaining $x^\star \in [lo, hi]$ from the bracket of Lemma 4.1:

$$
\begin{aligned}
&(lo,\ hi) \leftarrow (0,\ x_{\max} - 1) \\
&\textbf{while } hi > lo: \\
&\quad m \leftarrow lo + \lceil (hi - lo)/2 \rceil \\
&\quad v \leftarrow E(m) \qquad \text{(one probe; } G(m) = m + v\text{)} \\
&\quad \textbf{if } v \geq 0:\ lo \leftarrow m + v \quad \textbf{else}:\ hi \leftarrow m + v \\
&\textbf{return } lo
\end{aligned}
$$

Proposition 5.2 (value reuse is free)**.** *The loop maintains $x^\star \in [lo, hi]$, terminates in at most K probes — the same worst case as Section 5.1 — and returns the bit-identical $x^\star$, including the largest-integer selection on the $f = 1$ plateau (Lemma 4.3). The published rate, the committed checkpoint, and the obligations of Section 5.7 are all unchanged.*

Proof. The invariant holds initially by Bracketing (Section 4.2); each update replaces an endpoint by $G(m)$, which Lemma 5.1 places between that endpoint and $x^\star$ inclusive. Progress: $lo < m \leq hi$, so the $v \geq 0$ branch gives $lo' = m + v \geq m > lo$ and the $v < 0$ branch gives $hi' = m + v \leq m - 1 < hi$ — never worse than moving to the midpoint, so the $\lceil \log_2 \rceil$ bound is inherited and reuse can only beat it. At termination $lo = hi = x^\star$. On the $f = 1$ plateau the $v \geq 0$ branch pushes lo upward through the block ($lo' = G(m) \geq m$), so the returned point is the largest integer with $E \geq 0$, the $x^\star$ of Lemma 4.1. The only change from Section 5.1 is the bracket-update rule. □

The gain is large because the recursion is locally inert wherever the junior buffer fully covers the senior delta ($\partial ST_{\text{eff}}/\partial JT_{\text{raw}} = 0$): one value jump lands on the root, so healthy and stressed-but-covered states — including $f = 1$ and a binding clamp — solve in 2–3 probes. The loop is live only where the senior holds claims on the pool leg (IL recovery, cross-claim attribution, the yield split), and there each jump still shrinks the overshoot by the local gain, so even deep distress converges in ~ 20 probes against the ~ 64 of width-limited bisection. Since the solve runs on every swap, add, and remove, this is a 5–30× cut in solver gas on the hot path, at the cost of a single changed line and no new assumptions.

5.3 The search bracket and the expressible range

The solver uses the *conservation bracket* $\mathcal{B}$ of Lemma 4.1 as its search domain. Its endpoints bracket $x^\star$ by construction, so the search always returns it. Each probe evaluates the frozen-L valuation in closed form, not the pool's invariant solver, so no probe re-triggers the E-CLP domain check. That check runs once, when L is computed at solve start from the live balances.

The largest root $x^\star$ is the economically correct answer, but the deployed pool can only *quote* rates inside its price band. Write α and β for the lower and upper bounds on the senior rate $y = x/N$ *after* the mapping below. The expressible range is then

$$x \;\in\; [N\alpha,\; N\beta]. \tag{3}$$

These bounds are per-solve. The pool's configured band constrains the *rate-scaled* senior price, so it is recentered at the cached rate y_0, and its orientation depends on the token sort:

- if the senior share is token0 (price quoted as quote-per-senior), the senior-rate bounds are the configured bounds directly.
- if the senior share is token1 (price quoted as senior-per-quote), they are the reciprocals of the configured bounds, swapped, then recentered at y_0. At rates near 1 this reduces to the $[N/\beta,\ N/\alpha]$ form of earlier materials. The exact rate-scaling factor in this mapping is pending verification against the deployed pool's convention.

After solving, the kernel checks $x^\star \in [N\alpha, N\beta]$. When it holds, the kernel publishes $y^\star$ and operations proceed normally. When $x^\star$ falls outside, the rate is real but unquotable — and the kernel does *not* revert the sync. The solved root is still the honest mark: NAV conservation holds and P_A is a conservative lower bound whatever the band (below it the valuation is the all-senior extension of Section 5.5), so the accountant checkpoint commits at $y^\star$ as usual. What the kernel withholds is the *tradeable* rate. An unquotable rate would have the pool price against a value it cannot represent, so its secondary market is suspended: operations that consume a quote — swaps and unbalanced, single-sided liquidity — revert, while operations that do not — primary senior issuance and redemption, which touch only the senior tranche's exogenous assets, and proportional junior add and remove, which is balance-ratio math with no price — proceed against the committed checkpoint. Senior protection across this state is held by the shared accountant's coverage requirement, which gates junior withdrawals independently of the band; the band governs only what the pool can *quote*, never whether the market can *settle*. The out-of-band condition is thus a genuine economic signal — a senior loss larger than the pool's band can represent — and the response is to close the secondary market while keeping primary issuance and redemption live, not to freeze the market. Which operations read the published rate, and so which the suspension must gate, is an implementation obligation (Section 5.8).

5.4 Computing the bracket endpoint

The upper endpoint $x_{\max} = ST_{\text{raw}} + \lfloor f\, V_{\max} \rfloor + 1$ (Lemma 4.1) needs one pool-derived number: a conservative upper bound on the junior mark. Since $P_A(y) = \lfloor f\, \widehat{V}(y) \rfloor$ and $\widehat{V}$ is non-decreasing with $\widehat{V}' = \widehat{D} = \min(D, N)$ vanishing above the band, the mark is largest at the upper band edge,

$$\sup_y \widehat{V}(y) \;=\; \widehat{V}(\beta) \;\le\; V_{\max}.$$

The implementation uses this tighter $\widehat{V}(\beta)$: it is the exact supremum of the clamped valuation, whereas $V_{\max} = \sup_y V(y)$ over-marks by the area the clamp discards. The bracketing argument of Section 4.2 is untouched, since $P_A \le \lfloor f\,\widehat{V}(\beta)\rfloor \le \lfloor f\,V_{\max}\rfloor$ at every candidate. This subsection records how $\widehat{V}(\beta)$ is computed; the construction is pool-specific and given here for the Gyro E-CLP.

The threshold rate. $D(y)$, the pool's demand for senior shares (equivalently the senior reserve at the fair point), is non-increasing in y: it sits at its all-senior plateau $D(\alpha)$ below the band, falls across it, and reaches 0 at the all-quote corner β. The clamp $\min(D, N)$ binds exactly where $D > N$. If $D(\alpha) \le N$ the pool can never demand more senior shares than exist, the clamp never binds, and $\widehat{V}(\beta) = V_{\max}$. Otherwise there is a unique *threshold rate* y_N in the band with

$$D(y_N) \;=\; N,$$

the clamp binding on $[0, y_N]$ and releasing on $[y_N, \beta]$.

The corner value. Splitting $\widehat{V}(\beta) = \int_0^\beta \min(D, N)$ at y_N — the integrand is the constant N below it and $D = V'$ above it, which telescopes — gives the closed form of the Clamp obligation (Section 5.7) evaluated at the corner:

$$\widehat{V}(\beta) \;=\; \underbrace{N\,y_N}_{\text{clamped rectangle}} \;+\; \underbrace{\big(V(\beta) - V(y_N)\big)}_{\text{unclamped tail}} \;=\; V_{\max} - b_q(y_N), \tag{4}$$

where $b_q(y_N)$ is the quote reserve at the threshold point; the compact right-hand form uses the fair-point identity $V(y_N) = b_q(y_N) + y_N\,D(y_N) = b_q(y_N) + N\,y_N$. Read plainly: the conservative corner is the all-quote value minus the quote the pool already holds at the rate where its senior demand first meets the supply. The discarded difference $V_{\max} - \widehat{V}(\beta)$ is exactly the phantom-coverage gap of Section 3.3, the value resting on senior shares beyond the N that exist.

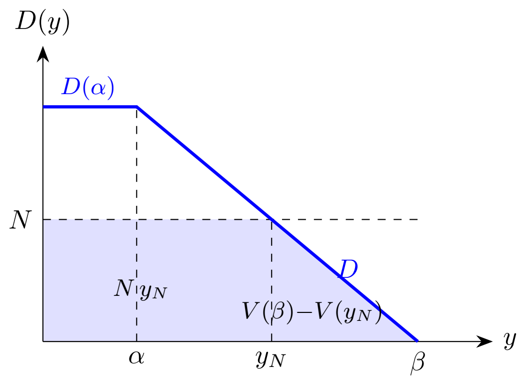

Figure 3: The bracket endpoint $\widehat{V}(\beta) = \int_0^\beta \min(D, N)$ is the shaded area under $\min(D, N)$: a rectangle of height N and width y_N where the clamp binds, plus the tail under D from y_N to the all-quote corner β. The unshaded wedge between D and the line N on $[0, y_N]$ — largest near $y = 0$, where E-CLP demand grows without bound — is the phantom coverage the clamp discards, exactly $V_{\max} - \widehat{V}(\beta)$.

Closed form on the E-CLP. Every quantity in Eq. (4) is a closed-form evaluation on the frozen curve, and the rate y_N need never be materialized. $V_{\max} = V(\beta)$ is the rate-scaled quote reserve at the all-quote corner — the curve's maximal balance for the quote leg (`maxBalances`) at the frozen invariant L — and the same corner quantity for the senior leg is the $D(\alpha)$ the guard above compares against N. The threshold's conjugate reserve $b_q(y_N)$ comes straight from the curve's reserve-given-reserve map: fixing the senior real reserve at N returns the quote reserve at the unique curve point where senior demand meets the supply (`calcYGivenX` for a senior token0, `calcXGivenY` for token1) — a closed-form ellipse inversion, no iteration. The endpoint is then $\widehat{V}(\beta) = V_{\max} - b_q(y_N)$; the rate y_N is just the spot price at that same point, needed only if the in-band check $y_N \in [\alpha, \beta]$ is performed, never for the value. Here N enters in the pool's rate-scaled balance units, the senior leg scaled at the frozen y_0. All evaluations run on the frozen curve the probes use (Section 3.3), so none re-triggers the pool's domain check.

Rounding. $\widehat{V}(\beta)$ is the one quantity rounded *up*. Every other rounding favours the senior tranche by rounding down (the floor in P_A, the invariant L, the published rate); an endpoint instead must certify a sign, so it may not underestimate the mark it bounds. Rounding $\widehat{V}(\beta)$ up — equivalently, one wei added to a rounded-down library output — keeps $x_{\max}$ a strict over-estimate and the bracketing of Lemma 4.1 intact.

5.5 Why the solve always converges

Lemma 4.1 is a statement about the function E. For it to govern the on-chain loop, three implementation invariants must hold. Each is cheap and is listed as an obligation in Section 5.7.

Frozen inputs. Every input to E (ST_{raw}, N, f, the invariant L, the cached rate y_0, and the checkpoint) is read once at solve start and never re-read. Bisection compares E at different probes and assumes one fixed function. Any external reads inside F (the yield-model view, the block timestamp) are constant across the bisection, since it runs in one block against a frozen checkpoint, so no probe sees a different function.

Live supply. $N \geq 1$. The map divides by N, so the supply must be positive. This is a *Dusk obligation*, not a property inherited from the shared tranche. The Dawn senior tranche mints shares only on deposit and burns them on redemption, seeds nothing at market creation, and applies no virtual-share offset in its conversion (at zero supply it converts one-to-one), so its supply can fall back to zero. A Dusk market must therefore pin $N \geq 1$ for the life of the market by construction, for instance a non-redeemable minimum senior position seeded at creation or a virtual-share offset in the senior conversion, so the division is always defined. The same floor is what discharges the $N \geq 1$ hypothesis of Lemma 4.1.

Total E. Every probe returns a value rather than reverting. E has two parts. P_A is the closed-form frozen-curve valuation, total on $\mathcal{B}$ by construction: below the band it is the all-senior line, above it the constant $V_{\max}$, and the E-CLP domain check runs only once, at L-computation (§5.3). F reverts only on a checkpoint that violates NAV conservation, and every stored checkpoint, the genesis checkpoint included, conserves by construction (Section 1.2). Its one external read, the yield-model view, must be a non-reverting view function (an obligation below). So on the reachable state space E is defined at every probe in $\mathcal{B}$.

Under these three invariants, the solve terminates in exactly K steps and returns the largest root $x^\star$. The only remaining reverts are external to the solver: a paused market, a stale or reverting price oracle, or the surrounding pool operation hitting its own Balancer limits. None of these is a failure of the solve, and in each case the cached rate is left untouched. Two guarantees follow, and only these two: the failed operation reverts before it can *consume* a stale rate, and a failed sync never *overwrites* the cache. Ordinary view reads (`getRate`) still return the last published rate: staleness is prevented from propagating through an operation, not from being read.

5.6 Iteration count and rounding

Two policy choices fix the solver's behaviour:

Iteration count is the K of Section 5.1, hardcoded per market to cover the largest reachable bracket, $2^K \geq x_{\max}+1$. Each step of E lies in $\{-1, 0\}$ (Section 4.1), so K halvings collapse every reachable bracket to the single integer boundary $x^\star$. No residual or iteration-count check is needed.

Rounding sends the published $y^\star = x^\star/N$ toward the senior tranche ($x^\star$ itself is an exact integer, Section 4.1). Together with the floor inside P_A (Eq. (1)) and the round-down of the invariant, every rounding step favours the protected tranche.

5.7 Implementation obligations

The correctness argument rests on the following, each verifiable per market. The first group makes E the right function, the second makes the solve converge, the third keeps every step conservative.

Senior independence. The senior raw NAV is marked from sources that do not read the pool (its BPT value, its spot price, or the published rate), so the loop has exactly one fixed-point variable.

Valuation wiring. The senior leg is lifted to WAD by its decimal factor 10^{18-d_s} (d_s its native decimals) and the published senior rate. The quote leg is lifted by 10^{18-d_q} and its rate q.

Frozen curve. L is computed once per solve from the balances scaled at y_0, rounded down (the underestimate), and held fixed across the bisection. The bracket endpoint is the clamped corner $\widehat{V}(\beta) \leq V_{\max}$ of Section 5.4, computed from this same frozen curve and rounded up.

Clamp. The valuation uses the integral form $\widehat{V}(y) = \int_0^y \min(D, N)$ (Section 3.3), with N read at solve start. In closed form this is Ny below the rate where demand falls to N, and V shifted by a constant above it (the split is unique when demand is monotone, as for the Gyro E-CLP). Capping the share count inside the point value, $b_q + y\min(D, N)$, is a different function that is not N-Lipschitz and can break monotonicity.

Frozen inputs, live supply, total E. As in Section 5.5: read once, $N \geq 1$, conserving checkpoint.

Total probes. The closed-form valuation is defined for every $x \in \mathcal{B}$, including the band extensions below α (all senior) and above β (all quote), and the yield-model view is a non-reverting view function. P_A is evaluated at full precision in x, never through a quantized intermediate rate, so a unit step of x moves $f\widehat{V}$ by at most f, the bound Stage 1 of Section 4.1 assumes. A quantized rate batches many x-steps into one jump of the mark and breaks the step bound.

Non-negative mark. $ST_{\text{raw}} \geq 0$, the hypothesis Lemma 4.1 consumes at the lower endpoint (Section 4.2). A raw NAV marks deposited assets, so a conforming quoter cannot report a negative value. One that can must revert instead of returning it.

Conservative rounding. Senior-favouring at every step: the invariant down, the valuation floor, and the published rate toward senior.

5.8 Rate-freshness obligations

The solver publishes $y^\star$, and the surrounding pool must consume it freshly. These obligations are on the wiring between hook, kernel, and pool rather than on the solve, and are listed for the implementer.

Sync coverage. The hook registers all three before-operation callbacks (before-swap, before-add, and before-remove), so every standard operation triggers a sync. Recovery-mode removes are the one carve-out, listed below.

Reload ordering. The kernel syncs inside the matching before branch, and Balancer reloads balances and rates for that operation only after the hook returns, so the operation executes against the freshly published rate. The reload reads the Vault's own internal balances, so the hook must publish through a Vault entrypoint, since a raw token transfer into the pool is not seen by the reload.

Non-reverting view. `getRate` returns the cached $y^\star$ and must not revert merely because the rate looks stale.

Failure semantics. A failed sync reverts the operation and does not overwrite the cache, and ordinary view reads still return the last published rate (Section 5.5).

Unquotable rate. When the solved root falls outside the expressible range $[N\alpha, N\beta]$ (Section 5.3), the sync still commits the checkpoint at $y^\star$; the kernel withholds $y^\star$ as a tradeable rate and gates the pool's secondary market. Operations that consume a quote (swaps, unbalanced liquidity) revert; primary senior issuance and redemption and proportional junior add and remove proceed. The implementer must confirm which Balancer paths read the published rate — in particular that proportional removes settle on raw balances without it — and wire the suspension to match. The last quotable rate is left cached for view reads.

Recovery mode. Recovery-mode removes are a separate path, not bound by this contract.